

PolyAgora Strategy Manifold V1

Strategy Eigenfields, Admissibility Geometry, and Runtime Allocation

PolygonEye / PolyAgora CTO Specification

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1 Purpose

This note defines **PolyAgora Strategy Manifold V1**. The goal is to upgrade the current PolyAgora architecture from a regime-aware allocation engine into a runtime system over **strategy eigenfields**.

The key change is that strategies are no longer treated merely as signals, factors, or return streams. They are treated as semi-independent structural modes inside market space. Momentum, Carry, Defensive, Convexity, Low Volatility, Commodity Stress, and other AGUR strategies become admissible or inadmissible depending on the position of the market inside the Reference Polygon.

PolyAgora therefore does not compete with strategies such as Momentum. It governs when Momentum is structurally admissible, how much exposure it should receive, and how transitions between strategies should occur as market geometry evolves.

2 Existing PolyAgora Runtime Stack

The current PolyAgora architecture already contains the necessary primitives:

- (i) Reference Polygon
- (ii) Runtime State X_t
- (iii) Survival Matrix S_t
- (iv) Blocks / Local Stars
- (v) Zone Classification
- (vi) Driver Seat
- (vii) Allocation Engine

The Reference Polygon defines the structural space. The runtime state X_t locates the market inside that space. The Survival Matrix certifies admissibility. Blocks and Local Stars identify coherent strategy regions. Zones determine exposure intensity. The Driver Seat allows PM-level steering without violating structural discipline.

3 Reference Polygon

We retain the current five-dimensional Reference Polygon:

$$RP = (V, T, G, C, R)$$

where:

V = Volatility / risk pressure,
 T = Trend / directional persistence,
 G = Gold–Copper / macro stress bridge,
 C = Carry / credit and yield stability,
 R = Rates / duration and funding stress.

At each time t , the market is represented by:

$$X_t \in RP.$$

Thus:

$$X_t = (V_t, T_t, G_t, C_t, R_t).$$

X_t is not a regime label. It is a live coordinate inside the admissible market geometry.

4 Strategy Eigenfields

Let the strategy universe be:

$$\Phi = \{\phi_1, \phi_2, \dots, \phi_n\}.$$

Each ϕ_i is a strategy eigenfield. It represents a semi-independent structural tendency inside market space.

Examples:

Strategy	Eigenfield Interpretation	Core Sensitivity
Momentum	Persistence field	Trend continuation
Carry	Yield / stability field	Funding and credit stability
Low Volatility	Coherence stabilizer	Volatility suppression
Convexity	Stress asymmetry field	Tail / crisis response
Defensive	Capital preservation field	Boundary contraction
Commodity Stress	Inflation / macro stress field	Gold–Copper dislocation
VIX Boundary	Rupture detector	Volatility boundary states
Mean Reversion	Equilibrium restoration field	Dispersion reversal
Buffett	Valuation anchor field	Overextension / quality

Momentum is therefore added not as an external benchmark, but as:

$$\phi_{\text{MOM}} \in \Phi.$$

5 Market as Superposition of Strategy Eigenfields

The market state can be approximated as a time-varying superposition of active strategy eigenfields:

$$X_t \approx \sum_{i=1}^n \alpha_i(t) \phi_i,$$

where:

$$\begin{aligned} \phi_i &= \text{strategy eigenfield,} \\ \alpha_i(t) &= \text{runtime admissibility / dominance weight.} \end{aligned}$$

The purpose of PolyAgora is not to maximize $\alpha_i(t)$ mechanically. It is to determine which ϕ_i remain structurally admissible at X_t .

6 Admissibility Function

Define the admissibility of strategy ϕ_i at time t :

$$A_i(t) = A(\phi_i, X_t) \in [0, 1].$$

Interpretation:

$$A_i(t) = \begin{cases} 1 & \text{fully admissible,} \\ 0 & \text{inadmissible,} \\ (0, 1) & \text{partially admissible / buffer state.} \end{cases}$$

For Momentum:

$$A_{\text{MOM}}(t) = A(\phi_{\text{MOM}}, X_t).$$

Momentum is admissible when the Reference Polygon indicates sufficient trend persistence, manageable volatility, and no dominant rupture condition.

7 Strategy-Regime Definition

A regime is not a market label such as “bull”, “bear”, or “risk-on”.

A regime is a stable admissibility configuration across strategy eigenfields:

$$\mathcal{R}_k(t) = \{\phi_i \in \Phi : A_i(t) \geq \theta_A\}.$$

Thus, regimes are subsets of admissible strategy eigenfields.

Examples:

$$\begin{aligned} \mathcal{R}_{\text{Trend}} &= \{\phi_{\text{MOM}}, \phi_{\text{Carry}}, \phi_{\text{LowVol}}\}, \\ \mathcal{R}_{\text{Stress}} &= \{\phi_{\text{Convexity}}, \phi_{\text{Defensive}}, \phi_{\text{VIX}}\}, \\ \mathcal{R}_{\text{Rotation}} &= \{\phi_{\text{MeanRev}}, \phi_{\text{CommodityStress}}\}. \end{aligned}$$

8 Survival Matrix over Strategy Eigenfields

Define the Strategy Survival Matrix:

$$S_t \in [0, 1]^{n \times n}.$$

Each entry:

$$S_t(i, j)$$

measures the survivability of coexistence or transition between strategy eigenfields ϕ_i and ϕ_j .

Interpretation:

$$S_t(i, j) \approx 1 \Rightarrow \text{strategy } i \text{ can coexist with or transition to strategy } j,$$

$$S_t(i, j) \approx 0 \Rightarrow \text{transition or coexistence is structurally unstable.}$$

This is not merely a correlation matrix. It is a structural survivability matrix.

9 Dynamic Survival Matrix

Let S_0 be the prior structural survival matrix and R_t the rolling empirical survival estimate.

Then:

$$W_t = \lambda S_0 + (1 - \lambda) R_t,$$

where:

$$\lambda \in [0, 1].$$

W_t is the runtime survival matrix used by the allocation engine.

10 Blocks and Local Stars

A block is a coherent subset of strategy eigenfields:

$$B_m \subseteq \Phi.$$

The internal survival of block B_m is:

$$\Gamma_t(B_m) = \min_{\phi_i, \phi_j \in B_m} W_t(i, j).$$

The Local Star is the block with the strongest worst-case internal survival:

$$B_t^* = \arg \max_{B_m} \Gamma_t(B_m).$$

Thus, a Local Star is not the best-performing strategy. It is the most structurally coherent coalition of strategy eigenfields at time t .

11 Soft Membership

For each block B_m , define soft membership:

$$\mu_m(X_t) \in [0, 1].$$

This measures the degree to which the current market position belongs to the admissibility region of block B_m .

Final block score:

$$Q_m(t) = \mu_m(X_t) \cdot \Gamma_t(B_m).$$

The selected block is:

$$B_t^* = \arg \max_m Q_m(t).$$

12 Momentum as Internal Strategy Eigenfield

Momentum is now included as:

$$\phi_{\text{MOM}}.$$

Its admissibility depends on:

$$A_{\text{MOM}}(t) = f(T_t, V_t, G_t, C_t, R_t).$$

A simple first implementation:

$$A_{\text{MOM}}(t) = \sigma(a_T T_t - a_V V_t - a_G G_t + a_C C_t - a_R R_t),$$

where σ is a logistic squashing function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

Interpretation:

Momentum becomes more admissible when trend persistence is strong and volatility, macro stress, and rates stress are controlled.

13 Zone Logic

Zones are derived from structural coherence, boundary distance, and block stability.

Let:

$$Z_t \in \{1, 2, 3, 4\}.$$

Zone	Meaning	Allocation Behavior
1	Coherent regime	Full allocation into Local Star
2	Early transition	Reduced allocation / buffer mode
3	Degradation	Defensive scaling begins
4	Rupture	Capital preservation / cash / defensive block

Zone classification may be computed as:

$$Z_t = Z(\Gamma_t(B_t^*), \mu_{B_t^*}(X_t), d_{\partial RP}(X_t), \Delta Q_t),$$

where:

$$\begin{aligned} d_{\partial RP}(X_t) &= \text{distance to Reference Polygon boundary,} \\ \Delta Q_t &= \text{change in leading block score.} \end{aligned}$$

14 Driver Seat

The Driver Seat introduces PM-level steering parameters:

$$D_t = (\kappa_{\text{convexity}}, \kappa_{\text{carry}}, \kappa_{\text{boundary}}, \kappa_{\text{recovery}}, \kappa_{\text{defensive}}).$$

These parameters do not override the geometry. They deform allocation intensity inside structurally admissible regions.

Final strategy score:

$$\tilde{A}_i(t) = A_i(t) \cdot G_i(D_t) \cdot \mathbf{1}_{\phi_i \in B_t^*}.$$

where $G_i(D_t)$ is the driver-seat preference multiplier for strategy ϕ_i .

15 Runtime Allocation

Raw allocation score:

$$p_i(t) = \tilde{A}_i(t) \cdot q_i(t),$$

where $q_i(t)$ is the strategy's internal quality score.

Normalize:

$$w_i^{\text{raw}}(t) = \frac{p_i(t)}{\sum_j p_j(t)}.$$

Apply zone gross exposure scalar:

$$g(Z_t) = \begin{cases} 1.00 & Z_t = 1, \\ 0.60 & Z_t = 2, \\ 0.30 & Z_t = 3, \\ 0.00 \text{ to } 0.20 & Z_t = 4. \end{cases}$$

Final allocation:

$$w_i(t) = g(Z_t) \cdot w_i^{\text{raw}}(t).$$

Cash weight:

$$w_{\text{cash}}(t) = 1 - \sum_i w_i(t).$$

16 Implementation Steps

1. Extract all AGUR strategy return streams.
2. Add Momentum 12–1 as an internal strategy stream ϕ_{MOM} .
3. Define each strategy’s eigenfield category:
 - Momentum / persistence
 - Carry / yield stability
 - Defensive / preservation
 - Convexity / stress asymmetry
 - Low Vol / coherence
 - Commodity Stress / macro rupture
 - VIX Boundary / volatility rupture
4. Compute $X_t = (V_t, T_t, G_t, C_t, R_t)$ exogenously.
5. Compute admissibility $A_i(t)$ for each strategy.
6. Compute rolling survivability matrix R_t .
7. Blend R_t with prior matrix S_0 to obtain W_t .
8. Identify coherent blocks B_m .
9. Select Local Star B_t^* .
10. Apply zone classification.
11. Apply Driver Seat deformation.
12. Normalize allocations and compute final weights.

17 Key CTO Design Rule

Momentum must not be treated as an external competitor to PolyAgora.

Instead:

$$\phi_{\text{MOM}} \in \Phi.$$

PolyAgora governs:

1. when Momentum is admissible,
2. when Momentum should be scaled down,
3. when Momentum can coexist with Carry or Low Vol,
4. when Momentum becomes dangerous near boundary states,
5. when transition should occur toward Defensive or Convexity blocks.

18 Core Claim

PolyAgora is not a strategy. It is a runtime control layer over strategy space.

Strategies such as Momentum, Carry, Convexity, Defensive, and Low Volatility are eigenfields inside the Reference Polygon. Regimes emerge from stable admissibility configurations among these eigenfields. The Survival Matrix governs survivable coexistence and transition. Local Stars identify coherent strategy coalitions. Zones and the Driver Seat determine exposure intensity.

Thus:

$$\text{Strategy} \neq \text{PolyAgora}.$$

Rather:

$$\text{PolyAgora} = \text{Runtime Governance of Strategy Eigenfields}.$$

19 One-Line Summary

Momentum optimizes inside regimes. PolyAgora governs navigation across regimes.